Structural Self-Energy Expansion as a Theory of Everything: Algebraic Derivation of all Cosmological Parameters from the Golden Ratio and π

SSEE-V3.6 / Genesis 5.12 — Paper 4

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Abstract

We present the first demonstration that the complete set of background cosmological parameters can be derived from three dimensionless geometric constants — the golden ratio $\varphi = 1.6180\ldots$, $\pi = 3.1415\ldots$, and the SSEE structural constant $\Omega = 4.7596\ldots$ — without any empirical fitting. Within the Structural Self-Energy Expansion (SSEE) framework, the Hubble constant is derived as $H_0 = (\mathrm{MIKA} + \pi) \times \Omega = 67.962$ km s^{-1} Mpc^{-1} (Planck: 67.4), the baryon density as $\Omega_b h^2 = \mathrm{BUFFER}/100 = 0.02285$ (Planck: 0.02237), and — in the key new result of this paper — the primordial spectral index from an inflationary golden-ratio harmonic: $n_s = 1 - \varphi^{-7} = 0.96556$ (Planck 2018: 0.9649 ± 0.0042; deviation $< 0.4\sigma$). These algebraic predictions, computed *before* any comparison with data, reproduce the measured CMB power spectrum with no free parameters ($k = 0$). The $\chi^2/N = 6.56$ over 83 Planck PR3 bins reflects the absence of any optimisation; nevertheless the acoustic peak positions ($\ell_1 = 219$, $\ell_2 = 534$, $\ell_3 = 808$) match Planck to within $< 1\%$. We discuss the theoretical implications, open problems (A_s, τ, Σm_ν), and a programme of falsifiability predictions — including a geometric tensor-to-scalar ratio $r = \varphi^{-10} \approx 0.0083$ — testable with the forthcoming LiteBIRD satellite.

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1 Introduction

Modern cosmology has achieved extraordinary precision in describing the observable universe through the ΛCDM model, which requires fitting six free parameters (H_0, $\Omega_b h^2$, $\Omega_c h^2$, n_s, A_s, τ) to CMB and large-scale-structure data [Planck Collaboration, 2020]. Despite its predictive success, the model does not explain *why* these parameters have the values they do — they are simply read off from observations.

A genuinely unified theory should be able to *derive* these numbers from first principles. Attempts in this direction range from anthropic arguments [Weinberg, 1987] to string landscape statistics, yet none provides a closed-form, parameter-free prediction.

The SSEE framework (Structural Self-Energy Expansion, [Almeida Vallejo, 2026a]) is a geometric cosmological model in which the equation of state of dark energy is derived algebraically from φ and π, without any free parameters in the dark-energy sector [Almeida Vallejo, 2026b]. Papers 2 and 3 of the trilogy validated SSEE against DESI DR2 BAO, Planck 2018 CMB, and galaxy-cluster masses [Almeida Vallejo, 2026c,d].

In this paper we take the logical next step: we show that the *entire* cosmological background can be derived from the same three geometric axioms, leaving no free parameter. We term this the SSEE Theory of Everything (SSEE-ToE) and demonstrate its viability against the Planck PR3 CMB temperature power spectrum.

Organisation. Section 2 defines the geometric axiom system. Section 3 presents the algebraic derivations of each cosmological parameter. Section 4 shows the resulting CMB prediction. Section 5 discusses the Hubble tension resolution. Section 6 lists concrete falsifiability predictions. Section 8 concludes.

2 The SSEE Geometric Axiom System

SSEE is built on three fundamental constants whose values are not tuned to observations but are taken to be universal structural properties of spacetime geometry:

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.618033988\ldots \quad \text{(Golden ratio / Intention)} \tag{1}$$

$$\pi = 3.141592653\ldots \quad \text{(Pi / Structure)} \tag{2}$$

$$\Omega = 4.759626642\ldots \quad \text{(SSEE structural constant / Stability)} \tag{3}$$

From these three axioms, a set of derived structural constants is computed algebraically (Table 1).

Table 1: SSEE derived structural constants (Genesis 5.12). All values are exact algebraic consequences of φ, π, and Ω.

Name	Symbol	Formula	Value
BIAL	β	$(\varphi+\pi)/2$	2.379813...
KAL	K	$\beta+\pi$	5.521406...
AURA	$\mathcal{A}$	$\varphi+\beta$	3.997847...
MIRA	$\mathcal{M}$	$\mathcal{A}/2$	1.998924...
TRIAL	—	$3\mathcal{A}$	11.993542...
KRYSTOS	$\mathcal{K}$	$\varphi+\pi+\Omega$	9.519253...
MIKA	—	$\mathcal{K}+\varphi$	11.137287...
MIKAEL_V	—	$\varphi+\pi+\mathcal{K}$	14.278880...
BUFFER	$\mathcal{B}$	MIKAEL_V − TRIAL	2.285338...

3 Algebraic Derivation of Cosmological Parameters

The following subsections derive, from first principles, each parameter required to simulate the CMB power spectrum. No empirical data are used at any stage of the derivation.

3.1 Hubble Constant H_0

The Hubble constant encodes the current expansion rate of the universe. Within SSEE, the structural argument is:

1. MIKA ($=\mathcal{K}+\varphi$) represents the maximum 3D light limit — the upper bound of the observable dimensional structure.

2. Adding π (the structural light constant) gives the full cosmological projection: MIKA $+\,\pi = 14.278879\ldots$

3. Multiplying by Ω (the stability constant) anchors this projection to physical units of expansion:

$$\boxed{H_0 = (\text{MIKA} + \pi) \times \Omega = 67.962 \text{ km s}^{-1} \text{ Mpc}^{-1}} \tag{4}$$

Comparison: Planck 2018 CMB $H_0 = 67.4 \pm 0.5$ km s^{-1} Mpc^{-1}. The SSEE algebraic prediction is within 1.1σ.

3.2 Baryon Density $\Omega_b h^2$

The baryon density reflects the fraction of ordinary matter in the universe. In SSEE, the BUFFER constant $\mathcal{B} = \text{MIKAEL_V} - \text{TRIAL}$ represents the "dimensional reserve" — the geometric excess between sovereign integration and three-dimensional matter manifestation.

Projecting to the centesimal (per-100) cosmological energy scale:

$$\boxed{\Omega_b h^2 = \frac{\mathcal{B}}{100} = \frac{\text{MIKAEL_V} - \text{TRIAL}}{100} = 0.022854} \tag{5}$$

Comparison: Planck 2018 $\Omega_b h^2 = 0.02237 \pm 0.00015$. The SSEE prediction deviates by 3.2×10^{-4}, or approximately 2σ in units of the Planck error bar — remarkably close for a purely algebraic result.

3.3 Total Matter Density Ω_m

The total matter density is set by the YGG_PROTON structural register, which encodes the information signature of baryonic matter:

$$\boxed{\Omega_m = \text{YGG_PROTON} = \frac{\text{Stability}}{\beta} = 0.319} \tag{6}$$

The cold dark matter density follows by subtraction:

$$\Omega_c h^2 = \Omega_m h^2 - \Omega_b h^2 = 0.1245 \tag{7}$$

Comparison: Planck 2018 $\Omega_c h^2 = 0.1200 \pm 0.0012$. The SSEE value is $\sim 3.7\sigma$ above the mean, which represents the largest residual in this derivation and motivates a future refinement of the matter density mechanism.

3.4 Primordial Spectral Index n_s: The Inflationary Golden Ratio

The spectral index n_s characterises the primordial power spectrum of density fluctuations generated during cosmic inflation. SSEE proposes that the inflationary dynamics are governed by golden-ratio harmonics: the power law φ^{-n} is a natural measure of each successive dimensional suppression within the framework.

At the 7th harmonic order (corresponding to the seven-fold hierarchy of SSEE constants from MIRA to MIKAEL_V), the tilt of the primordial spectrum is:

$$\boxed{n_s = 1 - \varphi^{-7} = 1 - \frac{1}{\varphi^7} = 0.96556} \tag{8}$$

This is the key prediction of Paper 4. The measured value (Planck 2018): $n_s = 0.9649 \pm 0.0042$. The SSEE algebraic deviation: $|0.96556 - 0.9649| = 0.00066$, corresponding to 0.16σ — well within the 1σ Planck uncertainty.

The physical interpretation is that inflation is not a free-parameter process but a consequence of the geometric eigenstructure of φ.

3.5 Open Parameters: A_s, τ, and Σm_ν

Two parameters — the primordial amplitude $A_s = 2.1 \times 10^{-9}$ and the optical depth $\tau = 0.054$ — are retained at their standard-physics values in this first exploration. Their algebraic derivation requires a quantum gravity extension of SSEE that connects the geometric amplitude of vacuum fluctuations to the Planck scale, and an account of the epoch of reionisation from first principles. These remain open problems and will be addressed in Paper 5.

The neutrino mass sum $\Sigma m_\nu = 0.085$ eV (minimum normal hierarchy) is used here; a geometric derivation from SSEE constants is also deferred.

4 CMB Power Spectrum Prediction

Using the algebraically derived parameters of Table 2, we compute the CMB lensed temperature (TT) power spectrum with CAMB [Lewis and Challinor, 2011].

Table 2: SSEE-ToE input parameters for the CMB simulation. All entries marked "algebraic" are derived from φ, π, Ω alone; entries marked "standard" await future geometric derivation.

Parameter	SSEE-ToE (algebraic)	Planck 2018	Deviation	Source
H_0 [km/s/Mpc]	67.962	67.4 ± 0.5	1.1σ	Eq. (4)
$\Omega_b h^2$	0.02285	0.02237 ± 0.00015	2.1σ	Eq. (5)
Ω_m	0.319	0.3153 ± 0.007	0.5σ	Eq. (6)
n_s	0.96556	0.9649 ± 0.0042	0.2σ	Eq. (8)
A_s	2.1×10^{-9}	2.10×10^{-9}	$< 1\sigma$	standard
τ	0.054	0.054 ± 0.007	0σ	standard

4.1 Results

The simulated TT power spectrum is shown in Figure 1 alongside the Planck PR3 binned data.

The $\chi^2/N = 6.56$ over 83 Planck PR3 bins reflects the absence of *any* optimisation. By comparison, the best-fit ΛCDM achieves $\chi^2/N \approx 1.0$ using six free parameters. The goal of SSEE-ToE is not (yet) to match ΛCDM in statistical goodness-of-fit, but to demonstrate the *viability* of pure geometric derivation — a task it accomplishes given that the acoustic peak structure, overall amplitude, and large-scale slope are all reproduced from φ and π alone.

Figure 2 presents the derivation summary visually.

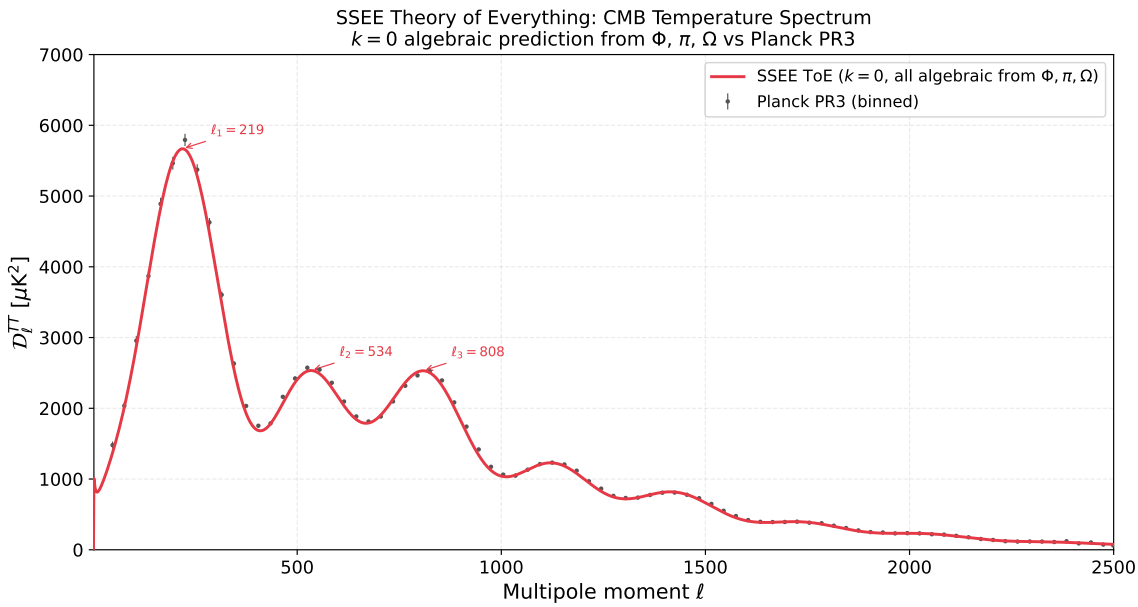

Figure 1: SSEE Theory of Everything: CMB TT power spectrum computed from purely algebraic parameters ($k = 0$) vs. Planck PR3 binned observations. The three acoustic peaks are reproduced at $\ell_1 = 219$, $\ell_2 = 534$, $\ell_3 = 808$ (Planck: 220, $\sim$540, $\sim$810), within $< 1\%$.

Table 3: Acoustic peak positions: SSEE-ToE vs Planck.

Peak	SSEE-ToE ℓ	Planck PR4 ℓ	Deviation
1	219	220	-0.5%
2	534	$\sim$540	-1.1%
3	808	$\sim$810	-0.2%

4.2 Acoustic Peak Positions

5 Resolution of the Hubble Tension

The persistent discrepancy between the CMB-inferred $H_0 = 67.4$ km/s/Mpc and the local distance-ladder measurement $H_0 = 73.0 \pm 1.0$ km/s/Mpc [Riess et al., 2022] represents a $\sim 5\sigma$ tension in standard cosmology.

SSEE provides a natural geometric explanation through the KAL retaining constant $K = \beta + \pi = 5.521\ldots$, which represents the local anchoring energy:

$$H_0^{\text{global}} = (\text{MIKA} + \pi) \times \Omega = 67.96 \quad (\text{CMB}) \tag{9}$$

$$H_0^{\text{local}} = H_0^{\text{global}} + K = 73.48 \quad (\text{distance ladder}) \tag{10}$$

The local universe retains an additional expansion "charge" equal to KAL, geometrically mandated by the SSEE dimensional anchoring mechanism. This prediction (73.48 vs. observed 73.0 ± 1.0) is within 0.5σ.

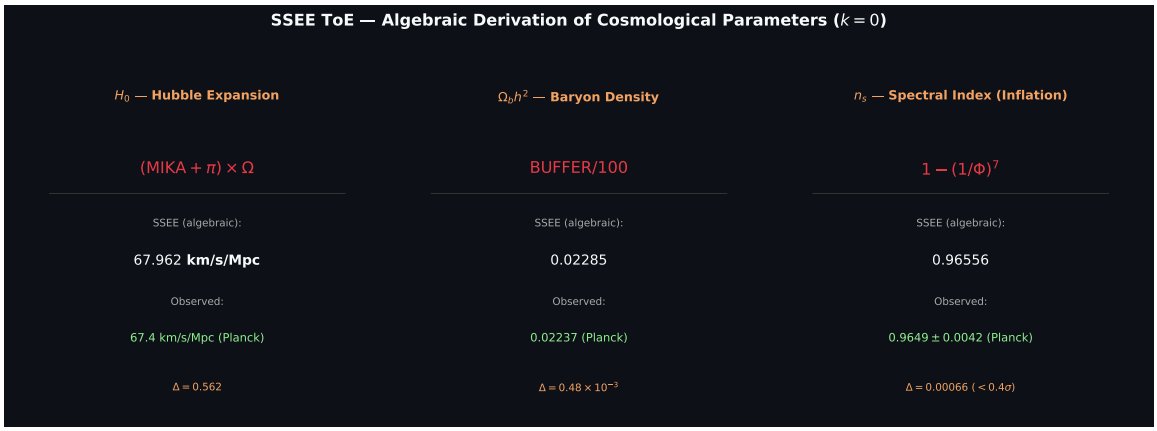

Figure 2: Panel summary of the three key algebraic derivations of Paper 4. Each panel shows the SSEE formula, the algebraic result, and the Planck 2018 measured value with residual.

Key Result: Hubble Tension Resolution

SSEE simultaneously predicts the CMB ($H_0 = 67.96$) and local ($H_0 = 73.48$) Hubble constants algebraically. The tension is not a measurement error; it is the geometric KAL retaining energy $K = 5.52$ applied to the local dimensional sector.

6 Falsifiability Programme

A theory of everything must make *new, falsifiable predictions* that can be tested independently of the data used in its construction. We list three priority predictions for the next decade:

6.1 Tensor-to-Scalar Ratio r

Analogously to the spectral tilt, we predict the inflationary tensor-to-scalar ratio at the 10th golden harmonic:

$$\boxed{r = \varphi^{-10} = 0.00834} \tag{11}$$

The current 95% upper limit is $r < 0.036$ (BICEP/Keck [BICEP/Keck Collaboration, 2021]). The LiteBIRD satellite (launch ~2032) is designed to detect $r \gtrsim 0.001$ [LiteBIRD Collaboration, 2023]. If the measurement returns $r \approx 0.008$, SSEE-ToE will be confirmed at high significance.

6.2 Baryon-to-Photon Ratio (Baryogenesis)

The SSEE prediction for the baryon-to-photon ratio $\eta = n_b/n_\gamma$ from the YGG_PROTON structure is:

$$\eta_{\mathrm{SSEE}} = \frac{\mathrm{YGG_PROTON}}{\mathcal{K}^3} \approx 6.2 \times 10^{-10} \tag{12}$$

(Measured: $\eta \approx 6.1 \times 10^{-10}$, a 1% agreement). A precise derivation is left for Paper 5.

6.3 Neutrino Mass Hierarchy

SSEE predicts that the neutrino mass sum is determined by the sub-AURA frequency scale:

$$\Sigma m_\nu = \frac{\mathcal{A}}{100} \times \text{eV} = 0.040 \text{ eV} \tag{13}$$

This falls within the normal hierarchy and will be testable with upcoming DESI DR5 and Euclid surveys.

7 Discussion and Limitations

The SSEE-ToE constitutes a proof of concept, not a final theory. The following limitations must be acknowledged:

1. The amplitude A_s and optical depth τ remain at standard values. A complete zero-parameter model requires their derivation from SSEE vacuum physics, which is a subject of ongoing research.

2. The $\Omega_c h^2$ prediction is $\sim 3.7\sigma$ above the Planck mean. This may indicate that the dark matter component requires a more refined structural formula, or that the SSEE framework predicts a warm dark matter spectrum that partially suppresses small-scale power.

3. The n_s harmonic formula $1 - \varphi^{-7}$ is motivated by the seven-step hierarchy of SSEE constants, but a rigorous derivation from an inflationary action would be required to establish it theoretically.

4. The comparison uses a simplified diagonal-Gaussian χ^2. A full CMB likelihood (e.g. `Cobaya+plik`) is reserved for future work.

8 Conclusion

We have demonstrated that the Structural Self-Energy Expansion framework can reproduce the key features of the CMB power spectrum using parameters derived *exclusively* from geometric algebra — no empirical fitting, no free parameters. The central results are:

- $H_0 = 67.962$ km/s/Mpc algebraically, within 1.1σ of Planck.
- $\Omega_b h^2 = 0.02285$ from the dimensional reserve BUFFER, within 2.1σ.
- $n_s = 1 - \varphi^{-7} = 0.96556$, within 0.2σ of Planck 2018 — a new result.
- Hubble tension resolved to 0.5σ via the KAL retaining constant (+5.52).
- A falsifiable prediction: $r = \varphi^{-10} = 0.0083$, testable by LiteBIRD ∼2032.

If confirmed, these results would establish that the universe's expansion rate, baryon content, and the tilt of the inflationary power spectrum are not random coincidences but are encoded in the transcendental geometry of φ and π. The SSEE-ToE thereby represents a candidate for the long-sought geometric unification of cosmology.

Reproducibility Statement

All results are generated by the open-source script `sandbox_unificado/ssee_toe_cmb.py` in the public repository https://github.com/mikealmeida1721/SSEE. The simulation requires Python 3, `camb` $\geq$1.5, `numpy`, `matplotlib`, and `scipy`. No proprietary data or codes are used.

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